The Ecology of Decisions: Probability Distortion as the Gap Between Bayesian and Ecological Rationality

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Abstract

Bayesian decision theory is widely used as the normative benchmark for rational behaviour, yet human choices often deviate from Bayesian prescriptions in systematic ways. Rather than interpreting these deviations as cognitive biases, this paper develops an ecological rationality framework in which behaviour is predicted directly from environmental structure. We show that any monotonic decision policy can be represented as a threshold applied to a distortion of Bayesian posterior probabilities. Using a Prelec weighting function to parametrise this distortion, we model optimal decision-making under uncertainty relative to geometric mean fitness. Through large-scale simulations of flee-stay survival decisions across diverse ecological conditions, we demonstrate that the optimal probability distortion parameter in this scenario is a predictable function of key ecological variables, primarily loss asymmetry and cue precision. This mapping introduces a framework for predicting behaviour from ecological parameters without fitting psychological constructs, and reframes probability distortion as the consequence of adaptation to environmental constraints.

1 Bayesian versus Ecological Rationality

1.1 The dominance of Bayesian normative frameworks

Across the cognitive, behavioural, and computational sciences, Bayesian principles are widely treated as the normative standard for rational behaviour. In cognitive psychology, the programme of "rational analysis" (Anderson, 1990) evaluates cognitive systems against Bayesian optimality criteria, and Bayesian cognitive modelling treats human judgement as approximating ideal Bayesian inference (Griffiths et al., 2015). In perceptual neuroscience, the Bayesian brain hypothesis (Knill and Pouget, 2004; Friston, 2010) proposes that neural systems should represent uncertainty and update beliefs according to Bayesian principles. In motor control, optimal feedback control models assume that agents

should integrate sensory uncertainty in a Bayesian manner to generate optimal actions (Körding and Wolpert, 2004).

In economics and decision theory, expected utility theory (von Neumann and Morgenstern, 1944) and Savage's axioms (Savage, 1954) define rational choice as Bayesian belief updating and utility maximisation. In artificial intelligence, Bayesian reinforcement learning, Bayesian optimisation, and Bayesian neural networks (Ghahramani, 2015) treat Bayesian inference and Bayesian decision principles as the mathematically principled foundation for optimal action selection. Across these fields, Bayesian rationality functions not merely as a descriptive model but as a normative benchmark: the standard against which human and artificial agents are evaluated.

1.2 Systematic deviations from Bayesian norms

Despite the ubiquity of Bayesian normative frameworks, human behaviour systematically and predictably deviates from Bayesian prescriptions. Classic findings include base-rate neglect (Bar-Hillel, 1980), conservatism in belief updating (Edwards, 1968), and overreaction to diagnostic information (Griffin and Tversky, 1992). In perceptual and cognitive judgement, people exhibit probability miscalibration (Gigerenzer and Brighton, 2009), subadditivity (Tversky and Kahneman, 1992), and systematic misestimation of likelihood ratios (Green and Swets, 1966). In decision-making under risk, Prospect Theory (Kahneman and Tversky, 1979) and cumulative Prospect Theory (Tversky and Kahneman, 1992) show that humans overweight small probabilities and underweight large ones in a highly regular fashion. Related work documents loss aversion (Tversky and Kahneman, 1992), ambiguity aversion (Ellsberg, 1961), and nonlinear sensitivity to evidence (Griffin and Tversky, 1992). These deviations are not random noise but lawful, reproducible patterns.

The standard interpretation is that humans are biased or irrational relative to Bayesian norms. Probability weighting, loss aversion, and heuristic decision rules are treated as cognitive imperfections—departures from the mathematically correct solution. This interpretation implicitly assumes that Bayesian expected utility maximisation is the appropriate normative benchmark for evaluating human behaviour.

1.3 Ecological rationality

An alternative perspective, ecological rationality, challenges the assumption that Bayesian coherence is the correct normative standard. According to this view, organisms evolve decision strategies that are adapted to the statistical structure of their environments (Gigerenzer and Brighton, 2009; Todd and Gigerenzer, 2012). Heuristics such as "take-the-best" or "recognition-based inference" are not flawed approximations to Bayesian inference but efficient solutions tailored to the information structure of natural environments.

Under ecological rationality, the relevant normative criterion is not expected utility but evolutionary fitness. A behaviour that violates Bayesian axioms may

nonetheless maximise survival or reproductive success in a given ecology. This perspective reframes many so-called "biases" as adaptive responses to asymmetric costs, noisy cues, or limited learning opportunities.

1.4 Mapping from ecology to behaviour

Existing work in ecological rationality has focused on specific heuristics or specific environments. For example, risk-sensitive foraging models (McNamara and Houston, 1986) show that variance in resource availability can favour risk-prone or risk-averse strategies. Error Management Theory (Haselton and Buss, 2000) argues that asymmetric costs of false positives and false negatives can favour systematic biases in threat detection.

However, these models are typically domain-specific and rely on building psychological models of human behaviour. There is no unified framework that predicts how an agent should deviate from Bayesian norms.

1.5 Why predicting behaviour from ecology, not psychology, matters

Psychological models of bias are primarily descriptive: they characterise how people behave but do not explain why those behaviours arise. Parameters such as loss aversion, probability weighting, or risk preferences are typically fitted to behavioural data and treated as stable psychological traits. However, these parameters often vary across contexts, tasks, and environments (Stewart et al., 2006), limiting their predictive power.

In contrast, ecological parameters—such as cue reliability, threat prevalence, cost asymmetries, and temporal horizons—generalise across contexts and can be measured independently of behaviour. Predicting behaviour from ecology rather than psychology offers several advantages. First, it enables prediction in novel environments, such as military threat detection, medical triage, or financial risk assessment, where psychological parameters are unknown but ecological structure is measurable. Second, it provides possible insights into the mechanistic explanations for biases: overweighting small probabilities may arise because rare but catastrophic events dominate fitness outcomes.

1.6 This paper: predicting behaviour from ecology via optimal probability distortion

In this paper, we propose a general framework for predicting human behaviour from ecological structure. We show that any monotonic decision rule can be represented as Bayesian inference combined with a distortion of posterior probabilities. We utilise the Prelec probability weighting function (Prelec, 1998), widely used in behavioural economics, as it provides a convenient and empirically validated parametrisation of this distortion. We then show that the distortion parameter α corresponds to the ecologically optimal strategy: the value of α that maximises geometric mean fitness in a given environment.

We simulate scenarios where agents must decide whether to stay or flee given uncertainty about the presence of a threat. Using large-scale simulations across a wide range of ecologies, we demonstrate that the optimal distortion parameter α^* is highly predictable from two ecological variables: the loss ratio and cue precision. This mapping allows us to predict behaviour directly from ecological structure, without assuming any specific psychological biases or heuristics. Finally, we show that the distribution of α^* across ecologies aligns with empirical estimates of human probability distortion, suggesting that human biases may be adaptive responses to ecological constraints rather than cognitive flaws.

This framework offers a principled method for predicting behaviour in novel environments. In doing so, it reframes cognitive biases as ecologically rational adaptations.

2 Probability Distortion

2.1 All monotonic policies reduce to a decision threshold

A central observation in decision theory is that any monotonic policy for a binary choice can be expressed as a threshold rule. Consider an agent who must choose between two actions—for example, fleeing or staying—based on a posterior belief p that a threat is present. A policy is monotonic if the probability of choosing the action "flee" is non-decreasing in p, that is, if the agent flees at p, they should also flee for all probabilities greater than p. This class includes virtually all psychologically and ecologically plausible strategies: Bayesian decision rules, heuristics, signal detection rules, and the decision policies implied by Prospect Theory.

By the monotonicity of the decision function, there exists a unique threshold τ such that the agent flees if and only if $p > \tau$. This follows from standard results in monotone comparative statics and from the representation theorems for cumulative prospect theory (Prelec, 1998; Wakker, 2010). Thus, regardless of the internal computations or heuristics used by the agent, the behavioural output can be represented as a single crossing point in the posterior probability space. This observation provides the foundation for a unifying account of decision-making under uncertainty.

2.2 The Bayesian decision threshold

Consider, for example, a scenario where a threat may be present and an agent must decide whether to stay or flee. Under Bayesian decision theory, the agent chooses the action that minimises expected loss. Let L denote the loss incurred by staying during a threat (false negative loss), and let F denote the cost of fleeing when no threat is present (false positive loss). The expected cost of staying is pL, while the cost of fleeing is $(1-p)F$. The Bayesian decision rule is therefore:

$$pL > (1-p)F.$$

Rearranging yields the canonical Bayesian threshold:

$$p > \frac{F}{L+F}.$$

This threshold is independent of the cue distribution, the learning process, or the temporal horizon; it depends only on the relative costs of false negatives and false positives. The Bayesian agent flees whenever the posterior probability of threat exceeds this fixed ratio.

2.3 All policies can be written using probability distortion

Because all monotonic policies reduce to a threshold rule, and because the Bayesian threshold is a special case, any policy can be transformed to the Bayesian posterior probability. Specifically, for any threshold τ, there exists a strictly increasing function $w\colon [0,1] \to [0,1]$ such that:

$$w(\tau) = \frac{L}{L+F}$$

Thus, any threshold rule can be written in the form:

$$w(p) > \frac{F}{L+F}.$$

Rearranging yields:

$$w(p)L > (1-w(p))F.$$

This representation shows that all deviations from the Bayesian threshold can be captured by a distortion of posterior probabilities. The function $w(p)$ encodes how the agent transforms the Bayesian posterior before comparing it to the cost ratio. This transformation may arise from cognitive biases, heuristics, or ecological adaptations.

2.4 Probability distortion as a universal representation

The representation above implies that probability distortion is not merely a psychological curiosity but a mathematically universal way of expressing threshold shifts. Any monotonic deviation from the Bayesian rule corresponds to some distortion function $w(p)$. This includes all heuristics and other ecologically rational strategies that humans use to make decisions.

Probability distortion therefore provides a unifying language for describing behavioural deviations from Bayesian norms.

2.5 Prospect Theory: human distortion fits the Prelec function

Empirical research in behavioural economics has shown that human probability distortion is well described by the Prelec function (Prelec, 1998):

$$w(p) = \exp(-(-\ln p)^{\alpha}),$$

where α controls the curvature of the weighting function. When $\alpha < 1$, the function exhibits the characteristic inverse-S shape: overweighting of small probabilities and underweighting of large ones. When $\alpha > 1$, the pattern reverses. Extensive empirical work (Gonzalez and Wu, 1999; Wu and Gonzalez, 1996) has shown that this functional form captures human behaviour across a wide range of tasks, including risky choice, ambiguity, and perceptual decision-making.

The Prelec function is attractive not only empirically but also theoretically: it is the only function that satisfies both invariance under scaling of the log-odds and a form of subproportionality that matches human behaviour. This makes it a natural candidate for representing ecologically rational distortions.

If humans behave ecologically rationally rather than following Bayesian rationality, then the value of α that maximises evolutionary fitness in a given environment should correspond to the behaviour of an ecologically rational agent.

3 Simulation Framework

To evaluate how ecological structure shapes optimal probability distortion, and therefore optimal behaviour, we constructed a large-scale simulation in which agents repeatedly face a binary decision problem under uncertainty. On each encounter, the agent receives a noisy cue, updates its Bayesian beliefs about threat prevalence, distorts the resulting posterior using a Prelec weighting function, and chooses whether to flee or stay. Survival outcomes accumulate multiplicatively across time, so long-run fitness is defined as the geometric mean of realised survival probabilities. By varying ecological parameters across a wide range of values, we identify the distortion parameter α^* that maximises fitness in each environment.

3.1 Environmental structure

Each agent inhabits an environment in which a latent threat may or may not be present on each trial. The true threat state on trial t is a Bernoulli random variable:

$$\text{Threat}_t \sim \text{Bernoulli}(\pi),$$

where π is the true prevalence of threats in the environment.

On each trial, the agent observes a cue x_t drawn from a Gaussian distribution whose mean depends on the true threat state:

$$x_t \sim \begin{cases} \mathcal{N}(\mu_1, \sigma^2), & \text{if threat is present,} \\ \mathcal{N}(\mu_0, \sigma^2), & \text{if threat is absent.} \end{cases}$$

We fix $\mu_0 = 0$ and $\mu_1 = 1$ without loss of generality, so that σ directly controls cue reliability. Smaller σ corresponds to more informative cues; larger σ corresponds to noisier environments.

For each ecology, we simulate n independent agents, each experiencing T encounters. The horizon T varies across ecologies and determines the temporal depth over which survival accumulates.

3.2 Bayesian learning

Agents do not know the true threat prevalence π. Instead, they maintain Bayesian beliefs and update them based on experience. The agent's belief about threat prevalence is represented as a Beta distribution:

$$\pi \sim \text{Beta}(\alpha_\pi, \beta_\pi),$$

initialised such that the prior mean equals π and the prior strength is controlled by a parameter s:

$$\alpha_\pi = s \cdot \pi, \qquad \beta_\pi = s \cdot (1 - \pi).$$

Learning dynamics. On each trial, the agent updates its beliefs only if it chooses to stay and therefore observes the true threat state. If the agent flees, no information is gained and the belief remains unchanged. Let $\text{Stay}_t \in \{0, 1\}$ denote whether the agent stayed. The updates are:

$$\alpha_\pi \leftarrow \alpha_\pi + \text{Stay}_t \cdot \text{Threat}_t, \qquad \beta_\pi \leftarrow \beta_\pi + \text{Stay}_t \cdot (1 - \text{Threat}_t).$$

The posterior mean $\hat{\pi}_t = \alpha_\pi / (\alpha_\pi + \beta_\pi)$ is used for decision-making.

3.3 Posterior probability estimation

Given cue x_t and current belief $\hat{\pi}_t$, the agent computes the posterior probability of threat using Bayes' rule:

$$q_t = \frac{f_1(x_t)\, \hat{\pi}_t}{f_1(x_t)\, \hat{\pi}_t + f_0(x_t)\, (1 - \hat{\pi}_t)},$$

where f_1 and f_0 are Gaussian likelihoods with means μ_1 and μ_0 and shared variance σ^2.

3.4 Probability distortion

The agent does not act directly on the Bayesian posterior q_t. Instead, it applies a Prelec distortion:

$$w(q_t) = \exp(-(-\ln q_t)^{\alpha}),$$

where α determines the curvature of the weighting function.

- $\alpha = 1$ recovers Bayesian behaviour,
- $\alpha < 1$ produces overweighting of small probabilities,
- $\alpha > 1$ produces underweighting.

3.5 Decisions and realised survival

On each trial, the agent compares the expected survival from fleeing versus staying.

Let:

- $L \in (0, 1)$ be the fitness if the agent stays and a threat is present (a false negative),
- $F \in (0, 1)$ be the fitness if the agent flees when no threat is present (a false positive).

Expected survival if the agent stays:

$$S_{\text{stay}} = w(q_t)\, L + (1 - w(q_t)) \cdot 1.$$

Expected survival if the agent flees:

$$S_{\text{flee}} = w(q_t) \cdot 1 + (1 - w(q_t))\, F.$$

The agent flees if:

$$S_{\text{flee}} > S_{\text{stay}}.$$

The realised fitness on each trial is:

$$S_t = \begin{cases} 1 & \text{if correct,} \\ F & \text{if false positive,} \\ L & \text{if false negative,} \end{cases}$$

3.6 Geometric mean fitness

Because survival outcomes multiply across time, long-run fitness is given by the geometric mean of realised fitness:

$$W = \exp\left(\frac{1}{T}\sum_{t=1}^{T}\log S_t\right).$$

For each value of α, we compute the mean geometric fitness across all simulated lives:

$$W(\alpha) = \frac{1}{n}\sum_{i=1}^{n}\exp\left(\frac{1}{T}\sum_{t=1}^{T}\log S_{i,t}\right).$$

3.7 Why Geometric Mean Fitness is the Normative Objective

A crucial assumption in our framework is that agents maximise geometric mean fitness rather than expected (arithmetic mean) utility. This choice is not arbitrary but follows directly from the multiplicative nature of survival and reproduction across time. When outcomes compound—as in sequential survival, growth, or reproduction—long-run success is governed by the geometric mean of payoffs, not their arithmetic average (Lewontin and Cohen, 1969; Tuljapurkar, 1990). In such environments, rare but severe losses have a disproportionate impact because a single near-zero outcome can drastically reduce long-run fitness.

This principle has been widely established in evolutionary biology, where fitness in fluctuating environments is determined by the long-run growth rate of lineages, formally equivalent to maximising expected log fitness (Houston and McNamara, 1999). Related arguments appear in behavioural ecology and economics, including risk-sensitive foraging theory (McNamara and Houston, 1986) and the Kelly criterion in optimal gambling and investment (Kelly, 1956; MacLean et al., 2011). These frameworks converge on the same conclusion: when payoffs multiply over time, variance is intrinsically costly, and optimal strategies must account for downside risk in a nonlinear way. Thus, the use of geometric mean fitness becomes a normative requirement for decision-making in multiplicative environments.

3.8 Optimising the distortion parameter

For each ecology in our simulation, we evaluate fitness across a grid of 200 logarithmically spaced α values:

$$\alpha \in [10^{-0.7},\, 10^{0.7}],$$

and identify the optimal distortion:

$$\alpha^* = \arg\max_{\alpha} W(\alpha).$$

3.9 Varying ecological parameters

To explore how ecological structure shapes optimal distortion, we vary six ecological parameters using Latin Hypercube Sampling:

- threat prevalence $\pi \in [0.01, 0.3]$,
- survival under threat $L \in [0.05, 0.75]$,
- survival under false alarm $F \in [0.9, 0.999]$,
- cue noise $\sigma \in [0.2, 1.5]$,
- horizon $T \in [5, 50]$,
- prior strength $s \in [0.5, 10]$.

For each of the 5000 sampled ecologies, we compute α^* and the corresponding fitness. This produces a large dataset of ecologically optimal distortions across a broad range of environments, forming the basis for the analyses in Section 4.

4 Results

4.1 Distribution of Ecologically Optimal Distortion Parameters

Across the 5,000 ecological scenarios sampled, the evolved distortion parameters α^* exhibit a broad but highly structured distribution (Figure 1). The mean value is 0.87, with a median of 0.77 and a standard deviation of 0.59, indicating that optimal behaviour typically requires substantial deviation from linear Bayesian weighting ($\alpha = 1$). The distribution is skewed toward values below one, reflecting the prevalence of ecologies in which rare but catastrophic threats dominate long-run survival outcomes.

This pattern is consistent with the theoretical expectation that geometric mean fitness favours cautious overweighting of small probabilities when losses are multiplicative and asymmetric. Environments with high loss ratios or noisy cues disproportionately reward strategies that treat low-probability threats as more consequential than their Bayesian posterior would suggest. Conversely, values $\alpha^* > 1$ arise primarily in ecologies where fleeing is costly relative to the expected threat.

4.2 Predicting α^* from Ecological Variables

To quantify how strongly ecological structure determines the optimal degree of distortion, we trained a Random Forest regressor to predict α^* from the six underlying ecological parameters $\{\pi, L, F, \sigma, T, s\}$. The model achieved a cross-validated r^2 of 0.76, indicating that approximately three quarters of the variance in α^* is predictable from environmental structure alone.

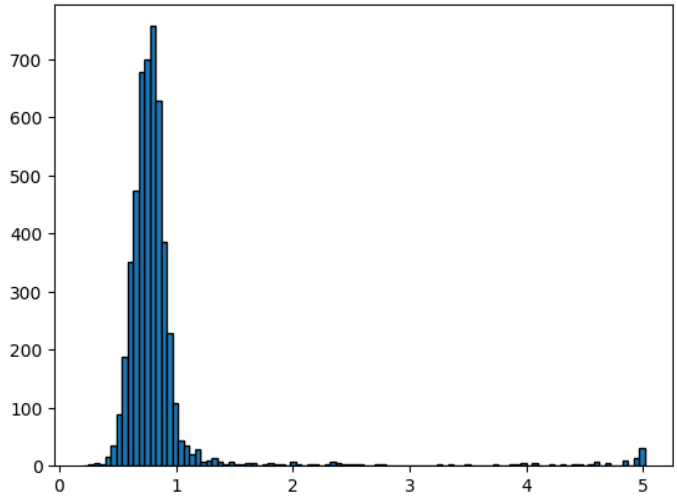

Figure 1: Distribution of evolved distortion parameters α^* across 5000 ecological parameter configurations. Values $\alpha^* < 1$ correspond to cautious overweighting of small probabilities, whereas $\alpha^* > 1$ corresponds to decisiveness or underweighting of small probabilities.

This level of predictability is notable given the stochastic nature of the simulation. Each estimate of α^* reflects the outcome of 3,000 independent survival trajectories, each influenced by noisy cues, finite horizons, and action-dependent learning. These sources of randomness introduce inherent variability into the mapping from ecological parameters to optimal distortion. That the Random Forest nonetheless explains the majority of variance demonstrates that α^* is not a fragile or idiosyncratic quantity: it is a stable function of ecological structure.

4.3 Three Composite Ecological Axes

Inspection of the Random Forest revealed that the six ecological parameters can be meaningfully summarised by three composite axes:

$$R = \frac{\pi \log L}{(1-\pi)\log F}, \tag{1}$$

$$P = \frac{1}{\sigma}, \tag{2}$$

$$H = \frac{T}{s}. \tag{3}$$

These axes capture 81% of the cross-validated variance in α^*, exceeding the predictive performance of the original six-parameter model. This improvement indicates that the ecological decision problem is effectively low-dimensional:

optimal distortion is governed primarily by relative losses and cue reliability, with horizon length playing a minor role.

Each composite has a clear ecological interpretation. The *Loss Ratio* R quantifies the asymmetry between false negatives and false positives in log-fitness space. The *Precision* axis P reflects the informativeness of environmental cues. The *Horizon* axis H represents the effective number of multiplicative survival episodes relative to prior strength.

4.4 Partial Dependence of α^* on Ecological Axes

Partial dependence curves reveal how each composite axis shapes the optimal distortion parameter.

Horizon. As shown in Figure 2, α^* decreases modestly with exponentially increasing horizon H. Longer effective horizons amplify the impact of rare catastrophic events on geometric mean fitness, making cautious overweighting of small probabilities more advantageous. The effect is relatively weak compared to the other axes, consistent with its low permutation importance.

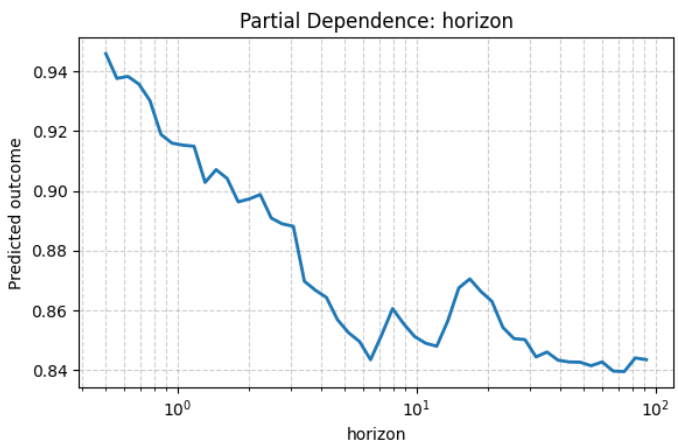

Figure 2: Partial dependence of α^* on Horizon ($H = T/s$). Longer horizons favour more cautious weighting (lower α^*).

Precision. Cue precision has a strong effect on α^* (Figure 3), which decreases as precision exponentially increases. This is because, as precision increases, posterior probabilities become more extreme, concentrating near 0 or 1. In such environments, α^* is driven downward in order to maintain equivalent levels of caution. The steepness of this decline highlights the importance of cue reliability in shaping adaptive decision strategies.

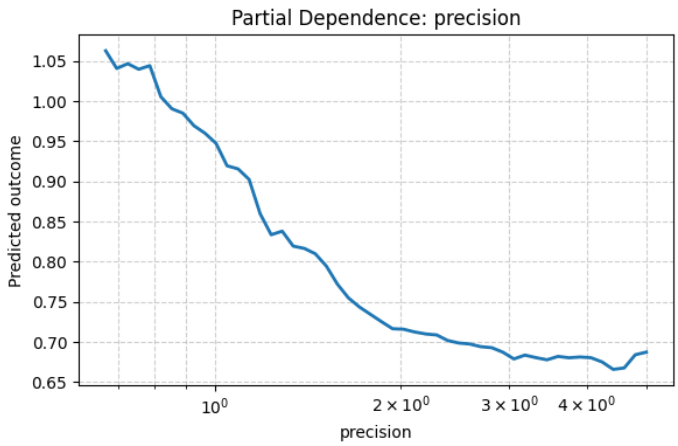

Figure 3: Partial dependence of α^* on Precision ($P = 1/\sigma$). Higher precision leads to stronger probability distortion (lower α^*).

Loss Ratio. The Loss Ratio R is the dominant determinant of α^* (Figure 4). At low R values, $\alpha^* > 1$, indicating that decisiveness is optimal when fleeing is costly. As R increases from very small values, α^* declines rapidly, reflecting the increasing asymmetry between the costs of false negatives and false positives. For large R, the curve flattens and then rises, indicating that less distortion is needed when fleeing is almost always favoured regardless of cue information. The minimum distortion is found when $R \approx 1$, which is when the expected costs of staying and fleeing are equal.

4.5 Permutation Importance

Permutation importance analysis confirms the primacy of the Loss Ratio in determining α^*. Shuffling R reduces predictive performance by 1.52 units of r^2, far more than the reductions observed for Precision (0.70) or Horizon (0.05). This ranking reflects the underlying structure of the decision problem: the asymmetry of costs (captured by R) is the primary driver of optimal behaviour, cue informativeness modulates the reliability of posterior estimates, and horizon length plays only a minor role.

Together, these results demonstrate that the magnitude and direction of probability distortion are lawful consequences of ecological structure. The evolved distortion parameter α^* is not arbitrary but reflects a principled adaptation to the statistical and energetic constraints of survival.

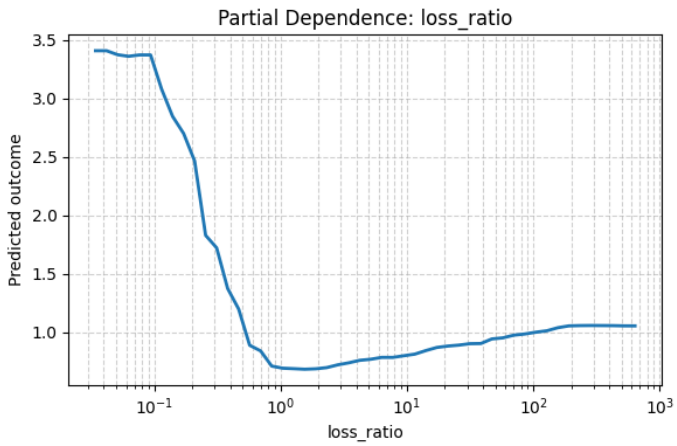

Figure 4: Partial dependence of α^* on Loss Ratio ($R = \frac{\pi \log L}{(1-\pi)\log F}$). Moderate-to-high loss ratios favour strong cautious probability distortion (low α^*), whereas very low ratios favour decisive distortion.

5 Optimal Distortion

5.1 Alignment Between α^* and Empirical Human Distortion

A striking outcome of the simulation is that the distribution of ecologically optimal distortion parameters α^* closely matches empirical estimates of human probability weighting. Across a wide range of decision-making tasks, human behaviour is well described by Prelec weighting functions with α values typically ranging from 0.55 to 0.95 (Prelec, 1998; Gonzalez and Wu, 1999; Wu and Gonzalez, 1996). The median value obtained in our simulations of 0.77, falls squarely within this empirically observed range.

This alignment suggests that the characteristic inverse-S shape of human probability weighting may not reflect a cognitive bias or computational limitation, but rather an adaptive response to the statistical structure of ancestral environments. In ecologies where rare but catastrophic events dominate long-run fitness, overweighting small probabilities is not a distortion but a necessity. The fact that the evolved distribution of α^* mirrors human behaviour across diverse tasks provides convergent evidence for the ecological rationality of probability distortion.

5.2 The Dominant Role of Loss Ratio

The Loss Ratio emerges as the primary determinant of the optimal distortion parameter. This ratio captures the expected severity of threats relative to the cost of fleeing, and therefore quantifies the asymmetry between false negatives

and false positives. As shown in the partial dependence analysis (Figure 4), α^* declines sharply as R increases from low to moderate values. When threats are rare or mild relative to the cost of fleeing, the optimal strategy is to underweight small probabilities (higher α^*), reflecting the fact that excessive caution is costly. As R increases, the cost of a false negative grows, and the optimal strategy shifts toward overweighting small probabilities (lower α^*). When R is high, α^* begins to rise back towards 1, as fleeing is almost always favourable anyway.

This pattern is consistent with the predictions of Error Management Theory (Haselton and Buss, 2000), which argues that asymmetric costs of errors favour systematic biases in decision-making. Our results extend this theory by quantifying the precise form of the optimal bias and showing that it corresponds to a specific distortion of posterior probabilities.

5.3 Cue Precision and the Amplification of Distortion

Cue precision exerts a strong influence on the optimal distortion parameter. As shown in Figure 3, α^* decreases steeply as precision increases. When cues are highly reliable, Bayesian posteriors become more extreme, clustering near 0 or 1. In such environments, large distortions of probability are needed to alter behaviour. To maintain an appropriate level of caution, the optimal strategy requires stronger curvature in the weighting function (lower α^*).

This result highlights a subtle but important point: ecological rationality depends not only on the structure of payoffs but also on the structure of information. When cues are noisy, posterior probabilities remain moderate, and only mild distortion is needed to achieve adaptive behaviour. When cues are precise, the agent must compensate for the extremity of Bayesian posteriors by applying stronger distortion. This interaction between information reliability and decision thresholds provides a mechanistic explanation for why probability weighting becomes more pronounced in perceptually or cognitively demanding tasks.

5.4 Horizon Effects and the Accumulation of Rare Events

The Horizon axis $H = T/s$ has only a minimal influence on the optimal distortion parameter. Permutation importance is extremely small, and removing H barely changes predictive performance. This insensitivity reflects the fact that learning plays little role in determining optimal distortion. Although agents update their beliefs about threat prevalence over time, these updates occur only when the agent stays and are typically small relative to the immediate decision pressures imposed by the Loss Ratio and cue Precision. As a result, variation in horizon length primarily affects the amount of data accumulated, but does not fundamentally alter the structure of the decision problem or the optimal mapping from posterior beliefs to actions. Each time step presents the same cue–posterior–action geometry, and geometric mean fitness already captures the multiplicative consequences of rare events even over short horizons. Consequently, optimal probability distortion is determined almost entirely by

ecological payoffs and information reliability, rather than by the dynamics of learning.

Despite its weak overall impact, the partial dependence curve shows a shallow downward trend: α^* decreases slightly as H increases. Longer horizons marginally amplify the consequences of rare catastrophic events, nudging the optimal strategy toward more cautious weighting of small probabilities. However, this modulation is secondary and does not reshape the decision boundary. Horizon length therefore plays only a minor role in determining α^*, consistent with its negligible contribution to predictive accuracy.

5.5 The Crucial Role of Geometric Mean Fitness

A key insight from the simulation is that the emergence of probability distortion depends critically on the use of geometric mean fitness as the objective function. When fitness is computed using arithmetic mean loss, the optimal distortion parameter rises to a mean of 1.02, with a median of 0.87, rising much closer to linear Bayesian weighting. This probably occurs because arithmetic averaging does not penalise variance or the multiplicative accumulation of rare catastrophic events. Under such an objective, the optimal strategy is to treat probabilities veridically.

In contrast, geometric mean fitness captures the multiplicative nature of survival and the disproportionate impact of rare but severe losses. Under this objective, the optimal strategy requires systematic distortion of posterior probabilities. This result provides a principled explanation for why Bayesian decision theory, which implicitly assumes additive utilities and arithmetic averaging, fails to capture adaptive behaviour in ecologically realistic environments.

5.6 Summary

Together, these findings demonstrate that the optimal degree of probability distortion is a predictable function of ecological structure. The evolved distortion parameter α^* aligns closely with empirical human behaviour, varies systematically with loss asymmetry and cue reliability, and primarily emerges when fitness is multiplicative. These results support the central claim of this paper: probability distortion is not a cognitive bias but the signature of ecological adaptation.

6 Implications

6.1 Bayesian Decisions Are Often Suboptimal in Ecological Contexts

The results presented here demonstrate that Bayesian decision-making—long treated as the normative ideal across cognitive science, economics, and artificial intelligence—is frequently suboptimal in ecologically realistic environments.

Bayesian thresholds assume additive utilities, linear weighting of probabilities, and decision-making that optimises arithmetic expected value. However, in environments where losses accumulate multiplicatively, where cues are noisy, and where the costs of false negatives and false positives are asymmetric, these assumptions fail.

Our simulations show that the Bayesian threshold $p > F/(L+F)$ is optimal only in a narrow subset of ecologies. In the majority of environments sampled, the optimal strategy requires systematic distortion of posterior probabilities. This finding suggests that many deviations from Bayesian behaviour are not errors but adaptive responses to ecological constraints.

6.2 Probability Distortion as an Ecological Adaptation

A central implication of our framework is that probability distortion should be understood not as a cognitive bias but as an ecological adaptation. The distortion parameter α quantifies the degree of transformation required for a Bayesian agent to behave optimally in a given environment. When $\alpha^* < 1$, the agent overweighting small probabilities is not misrepresenting the world but compensating for the disproportionate impact of rare catastrophic events on geometric mean fitness. When $\alpha^* > 1$, underweighting small probabilities reflects environments in which excessive caution is costly.

This perspective reframes classic behavioural phenomena. The inverse-S shape of human probability weighting, long interpreted as a violation of rationality (Tversky and Kahneman, 1992), emerges naturally from ecological pressures. Error Management Theory (Haselton and Buss, 2000) is subsumed as a special case: asymmetric costs favour systematic distortions, but the precise form of the distortion is determined by the full ecological structure, not by error asymmetry alone. Risk-sensitive foraging (McNamara and Houston, 1986) similarly emerges as a consequence of multiplicative fitness, with probability distortion providing the mechanistic link between environmental variance and adaptive behaviour.

6.3 A General Framework for Understanding Bias

Because any monotonic decision rule can be expressed as a threshold on distorted probabilities, our framework provides a general method for understanding the biases produced by ecologically rational strategies. Rather than requiring detailed knowledge of the underlying heuristic or learning rule, one can characterise the behaviour of an agent by the distortion function $w(p)$ that maps Bayesian posteriors to effective decision thresholds.

This perspective unifies diverse behavioural phenomena under a single mathematical framework. By expressing behaviours such as heuristics in terms of α^*, we obtain a principled, quantitative account of how ecological structure shapes decision-making, without needing to understand the underlying processes.

6.4 Predicting Behaviour from Ecology

One of the most powerful implications of this framework is that it enables prediction of behaviour directly from ecological structure. Given the loss ratio R and cue precision P, one can compute the optimal distortion parameter α^* and thereby predict the agent's decision thresholds, risk preferences, and sensitivity to rare events in the simulated flee-stay scenarios. This predictive capability does not require fitting psychological parameters or assuming specific heuristics; it follows directly from the ecological environment.

This approach offers a generative model of behaviour that generalises across contexts. In military threat detection, environments characterised by high loss ratios and noisy cues predict strong overweighting of small probabilities, reflecting the high cost of missed threats. In medical triage, asymmetric costs similarly favour conservative decision thresholds, prioritising the avoidance of false negatives. In financial risk-taking, by contrast, environments with low loss ratios and high cue precision predict underweighting of small probabilities, leading to more risk-tolerant behaviour. In public health, rare but catastrophic risks, such as pandemics, favour strong probability distortion even when Bayesian posteriors are low, reflecting the disproportionate impact of extreme events on long-run outcomes. These predictions arise not from psychological traits but from ecological structure.

7 Future Progression

7.1 Relaxing Simplifying Assumptions

The present framework demonstrates that optimal probability distortion emerges naturally from ecological structure, but several simplifying assumptions offer opportunities for deeper theoretical development. First, we assumed that the distortion parameter α remains fixed across all decisions within a lifetime. In reality, organisms may adaptively modulate their degree of caution based on recent experience, internal states, or changing environmental conditions. Allowing α to vary dynamically—either as a function of posterior uncertainty, cue volatility, or accumulated losses—could reveal richer forms of adaptive behaviour.

7.2 Beyond Flee–Stay Decisions

Although the flee–stay paradigm captures a fundamental class of survival decisions, many real-world problems involve richer action spaces. Foraging, mate choice, habitat selection, and social interactions often require choosing among multiple actions with different payoff structures and information requirements. Extending the framework to multi-action or continuous-action environments would test the generality of the distortion principle and may reveal new forms of ecological adaptation.

Sequential decision problems also merit attention. In environments where actions influence future states or information availability, optimal strategies may

involve exploration–exploitation trade-offs, temporal discounting, or strategic information gathering. Incorporating these complexities would connect ecological rationality with sequential decision-making frameworks in behavioural ecology and reinforcement learning.

7.3 Beyond the Prelec Function

The Prelec function provides an elegant and empirically validated parametrisation of probability distortion, but it is not the only possible form. Alternative weighting functions—including linear-in-log-odds, power-law distortions, or efficient-coding–derived transformations—may better capture optimal behaviour in specific ecologies. Identifying the conditions under which different functional forms are optimal would deepen our understanding of the computational principles underlying ecological rationality.

Moreover, the Prelec function is monotonic and smooth, but optimal distortions in some environments may be non-monotonic or piecewise. For example, environments with multimodal threat distributions or discontinuous payoff structures may favour more complex transformations. Exploring these possibilities would broaden the theoretical landscape and clarify the limits of single-parameter models.

7.4 Empirical Tests of Ecological Prediction

A key prediction of our framework is that ecological variables—not psychological traits—should determine the degree of probability distortion observed in human behaviour. This prediction can be tested experimentally by manipulating loss ratios, cue reliability, or horizon length in controlled laboratory tasks.

Such experiments would provide a direct test of the ecological rationality hypothesis and could reveal systematic relationships between environmental structure and human probability weighting. Field studies in domains such as finance, medicine, and military decision-making could further validate the ecological predictions in naturalistic settings.

7.5 Conclusion

This work provides a general framework for predicting behaviour from ecological structure. By showing that any monotonic decision rule can be expressed as a distortion of Bayesian posterior probabilities, and by demonstrating that the optimal distortion parameter α^* is a lawful function of ecological variables, we unify diverse behavioural phenomena under a single theoretical principle.

By reframing bias as the consequence of ecological adaptation, this framework challenges the dominance of Bayesian rationality as a universal normative benchmark and offers a principled method for predicting behaviour in novel environments. Future work extending, testing, and applying this framework promises to deepen our understanding of decision-making across species, contexts, and disciplines.

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